

SEA SENTRY

(FISHERMAN BORDER ALERT SYSTEM)

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PROBLEM STATEMENT:

The recurring detention of Tamil Nadu fishermen due to unintentional crossing of the International Maritime Boundary Line (IMBL) remains a major maritime safety and livelihood challenge. In July 2026, the Sri Lankan Navy detained nine Indian fishermen and seized their fishing vessel after they allegedly crossed the IMBL, highlighting that such incidents continue to occur. Although several researchers have proposed GPS- and IoT-based border alert systems, many remain at the prototype or research stage, depend on communication networks, or are not widely adopted by small-scale fishermen due to cost and practical limitations. This highlights the need for an affordable, reliable, standalone, and user-friendly border alert system. The proposed Sea Sentry aims to bridge this gap by developing a practical GPS-based solution that can assist fishermen in avoiding accidental border crossings and contribute towards safer fishing operations.

ABSTRACT:

Fishing is one of the primary sources of livelihood for coastal communities, but accidental crossing of international maritime boundaries remains a major concern. Due to the absence of an affordable and reliable navigation assistance system, fishermen often unknowingly enter restricted waters, leading to legal consequences, vessel seizure, and threats to their safety. Existing maritime navigation systems are expensive and may require continuous communication infrastructure, making them unsuitable for small-scale fishermen.

The proposed project, Sea Sentry – Fisherman Border Alert System, aims to develop a low-cost, GPS-based safety device that alerts fishermen before they cross the international maritime boundary. The system is built using an Arduino Mega 2560 as the controller and a

Neo-6M GPS module for real-time location tracking. Predefined latitude and longitude coordinates are used to establish a virtual maritime boundary. When the fishing boat approaches or reaches the restricted zone, the Arduino processes the GPS data and immediately activates a buzzer and LED to provide clear audio and visual warnings to the fishermen.

The proposed system is simple, portable, energy-efficient, and easy to install on small fishing boats. It operates independently without requiring internet connectivity, making it suitable for remote marine environments. By providing timely alerts, the system helps prevent accidental border violations, enhances fishermen's safety, and reduces the risk of legal disputes. The project demonstrates the practical application of embedded systems and GPS technology in addressing real-world Objectives

OBJECTIVES:

To develop a GPS-based maritime boundary monitoring system for fishing vessels.

To determine the real-time geographical position of the boat using the Neo-6M GPS module.

To detect predefined maritime boundary coordinates with high accuracy.

To provide immediate audio and visual warnings through a buzzer and LED when the vessel approaches the restricted zone.

To design a compact, low-cost, and energy-efficient embedded system using Arduino Mega 2560.

To minimize accidental international border violations by fishermen.

To ensure reliable operation without depending on internet or cellular connectivity.

To create a user-friendly prototype suitable for installation on small-scale fishing boats.

To enhance maritime safety through continuous location monitoring.

To provide a scalable platform for future integration with GSM, IoT, or emergency communication modules.

maritime safety challenges while offering an affordable solution for coastal communities.

SCOPE OF THE PROJECT:

The proposed Sea Sentry – Fisherman Border Alert System is designed to enhance the safety of fishermen by continuously monitoring the boat's location using the NEO-6M GPS module. The system processes real-time GPS coordinates through an Arduino Nano and compares them with predefined maritime boundary coordinates. When the boat approaches or crosses the specified boundary, the system immediately activates a buzzer and LED to warn the fishermen, helping them avoid unintentional border violations.

The project focuses on developing a compact, portable, low-cost, and energy-efficient device that can be easily installed on small fishing boats. Its simple hardware design makes it suitable for operation even by users with limited technical knowledge while requiring minimal maintenance.

The scope of this project extends beyond border alerts. The system can serve as a foundation for future smart marine safety applications by integrating technologies such as GSM/4G communication for automatic alert messages, SOS emergency buttons for distress signaling, weather monitoring sensors, OLED/LCD displays for live navigation information, solar-powered battery charging, wireless communication, and mobile application support for real-time tracking and notifications.

By combining affordability, reliability, and scalability, this project has the potential to improve marine safety, reduce accidental border crossings, support the livelihood of fishermen, and contribute to safer and more efficient fishing operations.

EXPECTED OUTCOMES:

- ❖ Accurate GPS-based detection of maritime border proximity.
- ❖ Instant audio and visual alerts to warn fishermen.
- ❖ Prevention of unintentional international border crossings.
- ❖ Enhanced safety and security for fishermen at sea.
- ❖ Reduced risk of detention and legal issues.
- ❖ Reliable real-time location tracking.
- ❖ Low-cost, portable, and energy-efficient prototype.
- ❖ Easy installation and operation on fishing boats.
- ❖ Improved confidence during deep-sea fishing.
- ❖ Scalable design for future enhancements, such as GSM or IoT integration.

Demonstration of a practical solution using embedded systems.

Promotion of safer and smarter fishing practices.

LITERATURE SURVEY:

S.NO	AUTHOR	CONTRIBUTION	LIMITATION
1.	P.Dhivyabharathi& Y.Jeyasingh (2016)	Developed a GPS and GSM based border alert system using Arduino to warn fisherman near maritime boundaries	Requires GSM network coverage and increases system cost
2.	Asif Iqbal Mulla ,Sushant K.J Prashanth Kumar H.R Abubakar Shameez(2016)	Introduced GPS based warning zones with LCD display for fishermen	Does not provide cloud monitoring or emergency communication .
3.	S.Ranjith,Shreyas K. Pradeep Kumar&R.Karthik (2017)	Proposed an automatic GPS-GSM border alert system with notification features	Depends on GSM connectivity and higher power consumption
4.	Nagaraj Balakrishnan,Reshmi S.,Arunkumar R.&M.S.Pradeeepraj(2018)	Developed a smart rescue system with GPS and IoT support	Internet connectivity is required for full functionality
5.	Kumar Harsh,Sarvesh Ranjan,Rishu Anand & Abhishek Kumar (2019)	Designed an Arduino and GPS-based border detection system for fishermen	Limited emergency response features

6.	Aakash S.,Arun Raja R.,Muthu Guru Madhavan S.&Rejini K. (2020)	Proposed border alert and message transmission to coastal authorities.	Relies on communication network availability
7.	Dr.Lijo Jacob Varghese,A.Sakthivel, J.K.Saran Kirthic,S.Sharan& Shri Pavithra (2022)	Combined GPS,GSM,obstacle detection and emergency communication in a maritime safety system	Uses multiple sensors and modules increasing cost and system complexity

EXISTING SOLUTION ANALYSIS:

Several GPS-based maritime safety systems have been developed to reduce accidental border crossings by fishermen. Most existing solutions use GPS and GSM technologies to monitor the boat's location and send alerts when it approaches an international maritime boundary. Some advanced systems also integrate IoT, cloud monitoring, and mobile applications for real-time tracking.

However, these existing systems have several limitations:

- Dependence on GSM or Internet connectivity, which is often unavailable in deep-sea regions.
- High implementation and maintenance costs due to multiple communication modules.
- Increased power consumption, reducing battery life during long fishing trips.
- Complex hardware and software architecture, making installation and maintenance difficult.
- Many systems are not affordable for small-scale fishermen.
- Some solutions require smartphones or external monitoring stations for operation.
- Limited standalone functionality when communication networks fail.
- Lack of a simple, low-cost warning mechanism suitable for rural fishing communities.

The above limitations indicate the need for a simple, reliable, and cost-effective border alert system that can operate independently without relying on GSM or Internet connectivity

The proposed **Sea Sentry – Fisherman Border Alert System** uses an Arduino Mega 2560, Neo-6M GPS module, buzzer, and LED to provide real-time border warnings. The system is designed to operate as a standalone device, making it affordable, portable, easy to use, and suitable for small fishing boats while reducing the risk of accidental border crossings.

GAP ANALYSIS:

Existing System	Identified Gap	Proposed Solution
GPS-based border alert systems are available.	Many systems rely on GSM or Internet connectivity, which is unreliable in deep-sea regions.	A standalone GPS-based system that works without Internet connectivity.
Commercial marine navigation systems provide accurate tracking.	They are expensive and not affordable for small-scale fishermen.	A low-cost solution using Arduino Mega 2560 and Neo-6M GPS module.
Some systems provide only text or display-based alerts.	Fishermen may not notice visual messages while operating the boat.	Provides instant buzzer and LED alerts for quick attention.
Existing systems often integrate multiple communication modules.	Increased hardware complexity, power consumption, and maintenance cost.	Simple hardware architecture with fewer components and easy maintenance.
IoT-based monitoring systems offer remote tracking.	Dependence on network availability limits their usability offshore.	Real-time local border detection without requiring network coverage.
Many research prototypes are difficult	Bulky design and high installation cost.	Compact, portable, and easy-to-install prototype suitable for small boats.

to install on small fishing boats.

Existing systems mainly focus on location tracking.

Several systems require continuous monitoring of a display.

Limited emphasis on immediate preventive warning before border crossing.

Visual monitoring alone may not be effective in harsh marine conditions.

Alerts fishermen before crossing the predefined maritime boundary.

Audio (buzzer) and visual (LED) alerts ensure immediate notification.

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